

Ali Mortada AlSadah

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EDUCATION

University of California San Diego (UCSD)

Graduated: July, 2026

Bachelors of Science, Computer Engineering

San Diego, California

- Provost Honors

Relevant coursework: Circuits and Systems, Digital/Analog circuits. Operating Systems principles and programming, Statistics, probability, data analysis and mathematical modeling. Software Engineering and Web Development.

PROJECTS & WORK EXPERIENCE

Electronic Circuit Development

April 2026 – June 2026

Design, Testing, Experimentation

San Diego, California

- Built various electronic circuits in the UCSD electrical engineering lab, created detailed 30+ page reports on each circuit.
 - Active high/low/band pass filters, voltage follower circuit, Integrator/Differentiator Circuits, Wein bridge oscillator.
- Applied mathematical concepts of control and stability theory in circuit design process to match practical specifications.
- Verified designs, calculations and report with senior Electrical Engineering Professor and researcher.

Viterbi Convolutional Encoder and Decoder

January 2026 – March 2026

HDL Programming, Logic Synthesis/Verification and Validation

San Diego, California

- Designed and implemented hardware Viterbi Convolutional Encoder and Decoder derived from mathematical algorithm.
- Synthesizable design, Generated RTL schematics, built experience using FPGA design tools Quartus Prime and Questa, and hardware description language System Verilog.

Software Development in Team Environment

April 2025 – June 2025

Front-end programmer, designer

San Diego, California

- Front-end programmer on a 10-person team. 8-week long software development challenge.
 - Built 2 production-ready web apps practicing Agile and using version control software. Team placed in the top 5 out of 30 competing teams of equal size.
 - Collaboratively worked alongside back-end programming team members and project lead. Coordinated with team members in daily standups. Brainstormed and debated features.
- Built and used CI/CD pipelines. Deployed 10+ features. Found and fixed multiple bugs. Ensured software compatibility and accessibility.

AWARDS & CERTIFICATES

KAUST Gifted Student Program (KGSP) Scholarship

March 2022-August 2024

Recipient

Jeddah, Saudi Arabia

- KGSP is a prestigious scholarship awarded by the King Abdullah University of Science & Technology (KAUST) a select group of Saudi students. Allowed me access and opportunity to learn from and connect with lead researchers and driven peers.

Data Collection and Processing with Python

September 2023

Certification

University of Michigan

Saudi Arabia Cultural Mission (SACM) Arruad Scholarship

August 2022

Recipient

Riyadh, Saudi Arabia

SKILLS

Programming Languages: Python, C, C++, Java, Verilog/System Verilog, MATLAB, HTML/CSS/JavaScript

Software: Linux, LTspice, Cadence Virtuoso, Pspice, AutoCAD, Quartus Prime, Altera Questa*, Git

Hardware: Microcontroller (Arduino), FPGA (Artix-7), Raspberry Pi, Function Generator, Oscilloscope, Logic Analyzer